

Brutal Practice Paper B51 - Physical & Chemical Changes, Algebraic Expressions, Forests, Integers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. An iron nail of mass 20 g is left out until it rusts completely, and rusting adds 30% to its mass. What is the mass of the rusted nail, and is rusting a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
 - (A) 23 g, chemical
 - (B) 26 g, physical
 - (C) 26 g, both kinds
 - (D) 20 g, chemical
 - (E) 6 g, chemical
 - (F) 26 g, chemical
 - (G) 14 g, chemical

2. If $x = 3$ and $y = -2$, find the value of the expression $2x^2 - 3xy + y^2$. [\[Algebraic Expressions\]](#)
 - (A) 40
 - (B) -40
 - (C) 4
 - (D) 58
 - (E) 0
 - (F) 16
 - (G) 22

3. The tallest trees of a forest spread their branches to form a leafy roof. This roof layer is called the canopy. If a canopy tree stands 32 m tall and the understorey trees reach only one quarter of that height, how tall are the understorey trees? [\[Forests\]](#)
 - (A) 6 m
 - (B) 12 m
 - (C) 8 m
 - (D) 128 m
 - (E) 4 m
 - (F) 64 m
 - (G) 16 m

4. A weather balloon reads -7 degrees C at the ground. It cools by 4 degrees C for every kilometre it rises and warms by 3 degrees C for every kilometre it later falls. After rising 5 km and then falling 2 km, what temperature does it read? [\[Integers\]](#)
 - (A) -9 deg C
 - (B) -15 deg C
 - (C) -21 deg C
 - (D) -35 deg C
 - (E) -13 deg C
 - (F) -27 deg C
 - (G) -41 deg C

5. When 12 g of magnesium ribbon is burnt in air it forms 20 g of white magnesium oxide. What mass of oxygen from the air combined with the magnesium? [\[Physical & Chemical Changes\]](#)
- (A) 20 g
 - (B) 8 g
 - (C) 6 g
 - (D) 4 g
 - (E) 16 g
 - (F) 12 g
 - (G) 32 g
6. Subtract $3a^2 - 5a + 2$ from $7a^2 - 2a - 6$. In the answer, what is the coefficient of a ? [\[Algebraic Expressions\]](#)
- (A) 4
 - (B) 3
 - (C) 5
 - (D) -7
 - (E) -2
 - (F) 7
 - (G) -3
7. Decomposers turn dead leaves on the forest floor into humus. If 6 kg of dead leaves fall on each square metre every year and decomposers break down five sixths of it, how much material is broken down per square metre? [\[Forests\]](#)
- (A) 7.2 kg
 - (B) 6 kg
 - (C) 1 kg
 - (D) 36 kg
 - (E) 5 kg
 - (F) 30 kg
8. A submarine sits at a depth of -180 m. It rises 25 m every minute for 4 minutes, then dives 40 m every minute for 2 minutes. What is its final depth? [\[Integers\]](#)
- (A) -160 m
 - (B) -260 m
 - (C) -240 m
 - (D) -220 m
 - (E) -100 m
 - (F) -80 m
9. Baking soda reacts with vinegar to release carbon dioxide. If 8.4 g of baking soda gives 4.4 g of carbon dioxide, how much carbon dioxide will 21 g of baking soda give at the same rate? [\[Physical & Chemical Changes\]](#)
- (A) 2.5 g
 - (B) 10 g
 - (C) 4.4 g
 - (D) 11 g
 - (E) 8.8 g
 - (F) 13.2 g
 - (G) 12.5 g

10. The perimeter of a rectangle is $10x + 6$ and its length is $3x + 4$. Find an expression for its breadth.

[Algebraic Expressions]

- (A) $2x + 1$
- (B) $8x + 2$
- (C) $2x - 7$
- (D) $2x - 1$
- (E) $4x - 1$
- (F) $x - 1$
- (G) $7x + 10$

11. In a forest of 1200 trees, three eighths are tall canopy trees, one third are shrubs, and the rest are herbs. How many herb plants are there? [Forests]

- (A) 350
- (B) 300
- (C) 400
- (D) 500
- (E) 450
- (F) 250
- (G) 800

12. A quiz gives +5 for each right answer, -3 for each wrong answer, and -1 for each skipped question. Riya gets 12 right, 7 wrong, and skips 6 of the 25 questions. What is her total score? [Integers]

- (A) 54
- (B) 33
- (C) 45
- (D) 39
- (E) 60
- (F) 27

13. A photographer notices silver chloride turns grey in sunlight. If 14.3 g of silver chloride breaks into 10.8 g of silver plus chlorine gas, what mass of chlorine gas escaped? [Physical & Chemical Changes]

- (A) 25.1 g
- (B) 4.5 g
- (C) 3.0 g
- (D) 3.5 g
- (E) 10.8 g
- (F) 14.3 g
- (G) 2.5 g

14. When the expression $5(2x - 3) - 2(x + 4)$ is fully simplified, what is its constant term? [Algebraic Expressions]

- (A) 8
- (B) 7
- (C) 23
- (D) -7
- (E) -11
- (F) -23

15. Forests soak up rainwater like a sponge and reduce floods. If trees in 1 hectare can absorb 40000 litres of rainwater, how many litres can a forest patch of 6.5 hectares absorb? [\[Forests\]](#)

- (A) 200000 litres
- (B) 240000 litres
- (C) 26000 litres
- (D) 46500 litres
- (E) 260000 litres
- (F) 2600000 litres

16. Two integers have a product of -84. One of them is 12. Find the other integer, and then find its difference when subtracted from 4. [\[Integers\]](#)

- (A) 18
- (B) 3
- (C) -11
- (D) -7
- (E) -3
- (F) 11

17. A salt pan worker knows 250 g of sea water contains 9 g of dissolved salt. If all the water is dried off from 2 kg of this sea water, what mass of salt remains, and is drying off the water a physical or chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 72 g, both
- (B) 720 g, physical
- (C) 72 g, chemical
- (D) 18 g, physical
- (E) 72 g, physical
- (F) 36 g, physical
- (G) 9 g, physical

18. Three times a number is increased by 7, and the whole result is then doubled. Taking the number as x , find the value of this expression when $x = 4$. [\[Algebraic Expressions\]](#)

- (A) 52
- (B) 24
- (C) 44
- (D) 19
- (E) 38
- (F) 26
- (G) 31

19. One large tree gives out about 120 kg of oxygen in a year. A class of 40 students each needs 250 kg of oxygen in a year. How many such trees are needed to supply the whole class for a year? [\[Forests\]](#)

- (A) 100 trees
- (B) 333 trees
- (C) 84 trees
- (D) 48 trees
- (E) 80 trees
- (F) 120 trees

- 20.** A lift starts 3 floors above the ground (floor +3). It goes down 9 floors, then up 12 floors, then down 5 floors. On which floor does it stop? (Ground is floor 0.) [\[Integers\]](#)
- (A) floor 0
 - (B) floor +11
 - (C) floor -1
 - (D) floor -3
 - (E) floor +3
 - (F) floor +1
 - (G) floor +5
- 21.** During respiration in the body, glucose reacts with 192 g of oxygen to give 264 g of carbon dioxide and some water. If 180 g of glucose was used, what mass of water is formed? [\[Physical & Chemical Changes\]](#)
- (A) 264 g
 - (B) 372 g
 - (C) 108 g
 - (D) 96 g
 - (E) 120 g
 - (F) 84 g
 - (G) 156 g
- 22.** Add the expressions $4p^2 - 3p + 5$ and $-2p^2 + 7p - 9$. What is the resulting expression? [\[Algebraic Expressions\]](#)
- (A) $2p^2 + 10p - 4$
 - (B) $2p^2 + 4p - 4$
 - (C) $2p^2 + 4p + 14$
 - (D) $2p^2 + 4p - 14$
 - (E) $2p^2 - 4p - 4$
 - (F) $-2p^2 + 4p - 4$
- 23.** In the food chain grass to deer to tiger, only 10% of the energy passes to the next level. If the grass holds 10000 units of energy, how many units finally reach the tiger? [\[Forests\]](#)
- (A) 1 unit
 - (B) 50 units
 - (C) 100 units
 - (D) 8100 units
 - (E) 1000 units
 - (F) 10 units
- 24.** At midnight the temperature was -8°C . It then rose steadily to reach 16°C at noon, 12 hours later. By how many degrees did it rise on average each hour? [\[Integers\]](#)
- (A) 24°C per hour
 - (B) 4°C per hour
 - (C) 2°C per hour
 - (D) 3°C per hour
 - (E) 12°C per hour
 - (F) 8°C per hour

- 25.** A candle is burning. The wax near the flame melts while the wick burns away. If the 15 cm candle shortens by 2.5 cm every hour, how long will it take until only 2.5 cm of candle is left? [\[Physical & Chemical Changes\]](#)
- (A) 6 hours
 - (B) 10 hours
 - (C) 12 hours
 - (D) 3 hours
 - (E) 5 hours
 - (F) 4 hours
- 26.** A taxi charges a fixed 40 rupees plus 12 rupees for every kilometre travelled. For a ride of 8 km, how much of the total fare comes from the per-kilometre charge alone? [\[Algebraic Expressions\]](#)
- (A) Rs 40
 - (B) Rs 96
 - (C) Rs 80
 - (D) Rs 136
 - (E) Rs 56
 - (F) Rs 12
 - (G) Rs 104
- 27.** Trees release water vapour into the air through transpiration, which helps bring rain. If one tree gives off 200 litres of water on a hot day, how much water does a grove of 75 trees give off in a week of 7 such days? [\[Forests\]](#)
- (A) 1050000 litres
 - (B) 10500 litres
 - (C) 21000 litres
 - (D) 52500 litres
 - (E) 1400 litres
 - (F) 105000 litres
 - (G) 15000 litres
- 28.** An integer is multiplied by -4, and then 15 is added to the result, giving -13. What is the integer? [\[Integers\]](#)
- (A) -7
 - (B) 28
 - (C) 7
 - (D) -2
 - (E) 0
 - (F) -28
 - (G) 2
- 29.** A dairy sets 24 litres of milk into curd. Each litre of milk gives 850 g of curd. The souring of milk into curd is a chemical change. How many kilograms of curd are produced? [\[Physical & Chemical Changes\]](#)
- (A) 24 kg
 - (B) 2.04 kg
 - (C) 850 kg
 - (D) 20.4 kg
 - (E) 204 kg
 - (F) 18 kg
 - (G) 22.4 kg

30. Simplify $3(2a - b) - 2(a - 3b)$, then find its value when $a = 2$ and $b = -1$. [\[Algebraic Expressions\]](#)

- (A) 2
- (B) 5
- (C) 11
- (D) 8
- (E) 14
- (F) -5
- (G) -1

31. 100 kg of dead plant and animal matter on the forest floor forms about 18 kg of humus. If a forest floor collects 2500 kg of dead matter, how much humus will form from it? [\[Forests\]](#)

- (A) 45 kg
- (B) 360 kg
- (C) 450 kg
- (D) 540 kg
- (E) 4500 kg
- (F) 250 kg

32. A bank account starts at 0 rupees. There are 4 deposits of 250 rupees each and 6 withdrawals of 180 rupees each. What is the final balance? (Use a negative number if overdrawn.) [\[Integers\]](#)

- (A) Rs 2080
- (B) -Rs 1080
- (C) -Rs 80
- (D) -Rs 180
- (E) -Rs 2080
- (F) Rs 1000
- (G) Rs 80

33. When 23 g of sodium reacts completely with 35.5 g of chlorine, they form common salt with no leftover. What mass of salt is produced? [\[Physical & Chemical Changes\]](#)

- (A) 12.5 g
- (B) 23 g
- (C) 29 g
- (D) 35.5 g
- (E) 58.5 g
- (F) 46 g
- (G) 81.5 g

34. What expression must be added to $2x$ squared - $5x + 1$ to get $5x$ squared + $2x - 4$? [\[Algebraic Expressions\]](#)

- (A) $3x$ squared + $7x - 3$
- (B) $3x$ squared - $7x - 5$
- (C) $7x$ squared - $3x - 3$
- (D) $3x$ squared + $7x + 5$
- (E) $3x$ squared + $7x - 5$
- (F) $-3x$ squared + $7x - 5$

35. A deer reserve starts the year with 800 deer. Over the year the herd grows by 15%, but tigers eat 95 of the deer. How many deer are there at the end of the year? [\[Forests\]](#)

- (A) 815
- (B) 1015
- (C) 825
- (D) 785
- (E) 920
- (F) 735

36. Find the value of the product $(-2) \times (-3) \times (-5) \times (-1)$, and then add -4 to that product. [\[Integers\]](#)

- (A) 22
- (B) -30
- (C) 34
- (D) 26
- (E) -26
- (F) 30

37. 10 g of sugar is heated in a pan. It melts, turns brown, then black, leaving 4.2 g of black carbon behind while water vapour escapes. Is this a physical or a chemical change, and what mass escaped as vapour?

[\[Physical & Chemical Changes\]](#)

- (A) physical, 4.2 g
- (B) chemical, 4.2 g
- (C) chemical, 6.0 g
- (D) chemical, 5.8 g
- (E) physical, 5.8 g
- (F) chemical, 14.2 g
- (G) chemical, 5.2 g

38. Each side of an equilateral triangle is $2x + 3$ units long. Find its perimeter and then its value when $x =$

5. [\[Algebraic Expressions\]](#)

- (A) 26
- (B) 39
- (C) 45
- (D) 18
- (E) 36
- (F) 30
- (G) 13

39. Forests are called the green lungs of the Earth because they take in carbon dioxide and give out oxygen. If each tree absorbs 22 kg of carbon dioxide a year, how much do 350 trees absorb in 2 years?

[\[Forests\]](#)

- (A) 15400 kg
- (B) 770 kg
- (C) 7700 kg
- (D) 7000 kg
- (E) 154000 kg
- (F) 30800 kg
- (G) 1540 kg

40. The sum of three consecutive integers is -33. What is the largest of the three integers? [\[Integers\]](#)

- (A) -11
- (B) -10
- (C) -9
- (D) -13
- (E) -33
- (F) -12
- (G) -8

41. Electrolysis splits water into hydrogen and oxygen. From 18 g of water, 2 g of hydrogen is released. How much oxygen is released from 90 g of water at the same rate? [\[Physical & Chemical Changes\]](#)

- (A) 88 g
- (B) 16 g
- (C) 80 g
- (D) 90 g
- (E) 72 g
- (F) 10 g
- (G) 8 g

42. Find the value of the expression $a^2 - 2ab + b^2$ when $a = 7$ and $b = 4$. [\[Algebraic Expressions\]](#)

- (A) 25
- (B) 65
- (C) 121
- (D) 9
- (E) 33
- (F) 49
- (G) 81

43. Some birds eat berries and drop the seeds far away, helping the forest spread. If a bird scatters 40 seeds a day and only 5% of them grow into new plants, how many new plants come from 30 days of scattering? [\[Forests\]](#)

- (A) 1200 plants
- (B) 200 plants
- (C) 600 plants
- (D) 6 plants
- (E) 60 plants
- (F) 1140 plants

44. Divide -144 by -6, and then subtract the result from 10. [\[Integers\]](#)

- (A) -34
- (B) -14
- (C) 24
- (D) 34
- (E) -10
- (F) -24
- (G) 14

45. Quicklime reacts with water to form slaked lime, giving out heat. If 56 g of quicklime reacts with 18 g of water and nothing escapes, what mass of slaked lime is formed? [\[Physical & Chemical Changes\]](#)

- (A) 38 g
- (B) 18 g
- (C) 48 g
- (D) 56 g
- (E) 92 g
- (F) 74 g
- (G) 64 g

46. A craft pattern uses $4n + 1$ tiles in figure number n . How many more tiles are in figure 10 than in figure 6? [\[Algebraic Expressions\]](#)

- (A) 20
- (B) 24
- (C) 4
- (D) 66
- (E) 16
- (F) 40
- (G) 12

47. On bare land about 25 tonnes of soil per hectare wash away each year. Where a forest covers the land, this loss drops to only one fifth as much. How many tonnes of soil per hectare are saved by the forest each year? [\[Forests\]](#)

- (A) 15 tonnes
- (B) 4 tonnes
- (C) 21 tonnes
- (D) 5 tonnes
- (E) 25 tonnes
- (F) 30 tonnes
- (G) 20 tonnes

48. One frosty morning, five towns report temperatures of -3, 7, -8, 0 and -1 degrees C. What is the difference between the highest and the lowest temperature? [\[Integers\]](#)

- (A) 7 deg C
- (B) 15 deg C
- (C) 11 deg C
- (D) 5 deg C
- (E) -15 deg C
- (F) 1 deg C
- (G) 8 deg C

- 49.** A firework explodes, giving out bright light, a loud bang, a lot of heat, and clouds of gas. Counting light, sound, heat and the new gases, how many signs of a chemical change are shown here? [\[Physical & Chemical Changes\]](#)
- (A) 6 signs
 - (B) no signs
 - (C) 2 signs
 - (D) 1 sign
 - (E) 4 signs
 - (F) 5 signs
- 50.** Simplify the expression $7m - [3m - (2m - 5)]$ and find its value when $m = 6$. [\[Algebraic Expressions\]](#)
- (A) 67
 - (B) 19
 - (C) 41
 - (D) 25
 - (E) 31
 - (F) 6
 - (G) 37
- 51.** A forest canopy blocks 90% of the sunlight from reaching the ground. If 4000 units of light fall on the canopy, how much light reaches the forest floor, and which layer receives the least light? [\[Forests\]](#)
- (A) 400 units, the herb layer
 - (B) 3600 units, the herb layer
 - (C) 400 units, the shrub layer
 - (D) 40 units, the herb layer
 - (E) 400 units, the canopy
 - (F) 3600 units, the canopy
- 52.** A spider climbs a well that is 11 m deep. Each day it climbs 5 m up, but each night it slips 3 m back down. On which day does it first reach the top and get out? [\[Integers\]](#)
- (A) day 11
 - (B) day 7
 - (C) day 2
 - (D) day 4
 - (E) day 6
 - (F) day 3
- 53.** A silver spoon of mass 40 g slowly tarnishes, and the black coating that forms adds 1.5% to its mass. Tarnishing is a slow chemical change. What is the new mass of the spoon? [\[Physical & Chemical Changes\]](#)
- (A) 41.5 g
 - (B) 40.15 g
 - (C) 40.6 g
 - (D) 60 g
 - (E) 46 g
 - (F) 40.06 g
 - (G) 41 g

54. If $P = 3x + 2$ and $Q = x - 5$, find the simplified expression for $P - 2Q$. [\[Algebraic Expressions\]](#)

- (A) $x + 7$
- (B) $5x - 8$
- (C) $x + 2$
- (D) $x - 8$
- (E) $5x + 12$
- (F) $x + 12$
- (G) $x - 12$

55. A forest of 8 hectares stores about 50 tonnes of carbon in each hectare. If a fire destroys 3 of these hectares, how many tonnes of stored carbon are lost to the air? [\[Forests\]](#)

- (A) 50 tonnes
- (B) 150 tonnes
- (C) 300 tonnes
- (D) 250 tonnes
- (E) 100 tonnes
- (F) 400 tonnes
- (G) 16 tonnes

56. Two integers differ by 9, and their sum is -5. What is the product of these two integers? [\[Integers\]](#)

- (A) -5
- (B) 45
- (C) 9
- (D) -14
- (E) -45
- (F) -2

57. A 200 g block of ice melts completely into water, and then one quarter of that water boils away. Melting and boiling are physical changes. What mass of liquid water is left? [\[Physical & Chemical Changes\]](#)

- (A) 200 g
- (B) 160 g
- (C) 100 g
- (D) 50 g
- (E) 175 g
- (F) 150 g
- (G) 125 g

58. The sum of three expressions is $9x$. Two of them are $2x + 4$ and $4x - 7$. Find the third expression.

[\[Algebraic Expressions\]](#)

- (A) $3x + 10$
- (B) $x + 3$
- (C) $15x - 3$
- (D) $3x + 3$
- (E) $3x - 3$
- (F) $3x + 11$

- 59.** A forest guard plants saplings in 18 rows with 24 saplings in each row, but one sixth of all the saplings do not survive. How many saplings survive? [\[Forests\]](#)
- (A) 72
 - (B) 400
 - (C) 288
 - (D) 360
 - (E) 432
 - (F) 300
 - (G) 252
- 60.** A freezer is at -18 degrees C. While defrosting, it warms by 5 degrees C every 10 minutes. What temperature will it reach after 50 minutes? [\[Integers\]](#)
- (A) -43 deg C
 - (B) 2 deg C
 - (C) 32 deg C
 - (D) 12 deg C
 - (E) -7 deg C
 - (F) 7 deg C
 - (G) 25 deg C
- 61.** A grocer finds that 5% of his 480 mangoes have over-ripened, which is a chemical change. How many mangoes are still good to sell? [\[Physical & Chemical Changes\]](#)
- (A) 475 mangoes
 - (B) 460 mangoes
 - (C) 456 mangoes
 - (D) 24 mangoes
 - (E) 450 mangoes
 - (F) 432 mangoes
 - (G) 240 mangoes
- 62.** Evaluate the expression $5x - 3y + 2z$ when $x = -1$, $y = 2$ and $z = 4$. [\[Algebraic Expressions\]](#)
- (A) 19
 - (B) 3
 - (C) -7
 - (D) 1
 - (E) 7
 - (F) -3
 - (G) -19
- 63.** In a forest food web a tiger eats both deer and wild boar. If the tiger's yearly food of 500 kg is made up of 30% deer and 70% wild boar, how many kilograms of wild boar does it eat in a year? [\[Forests\]](#)
- (A) 350 kg
 - (B) 35 kg
 - (C) 500 kg
 - (D) 70 kg
 - (E) 300 kg
 - (F) 250 kg

- 64.** A share price begins the week at 200 rupees. It falls 12 rupees on Monday, falls 8 rupees on Tuesday, rises 15 rupees on Wednesday, and falls 5 rupees on Thursday. What is its value now? [\[Integers\]](#)
- (A) Rs 200
 - (B) Rs 210
 - (C) Rs 195
 - (D) Rs 190
 - (E) Rs 220
 - (F) Rs 185
 - (G) Rs 180
- 65.** A roof sheet has an area of 6 square metres and is coated with zinc to stop rusting, using 25 g of zinc per square metre. How much zinc is used, and what is the purpose of this coating? [\[Physical & Chemical Changes\]](#)
- (A) 125 g, to prevent rusting
 - (B) 300 g, to prevent rusting
 - (C) 31 g, to prevent rusting
 - (D) 150 g, to add weight
 - (E) 75 g, to prevent rusting
 - (F) 150 g, to start rusting
 - (G) 150 g, to prevent rusting
- 66.** Simplify $2a$ multiplied by $3a$ squared, and then add $4a$ cubed to the result. What is the final expression? [\[Algebraic Expressions\]](#)
- (A) $10a$ to the power 5
 - (B) $10a$ to the power 6
 - (C) $10a$ cubed
 - (D) $6a$ cubed (sum left undone)
 - (E) $24a$ cubed
 - (F) $6a$ to the power 6 + $4a$ cubed
- 67.** A patch of cleared forest needs about 60 years to fully grow back. If 5 such patches are cleared and left to recover one after another, one finishing before the next begins, how many years will it take to restore all 5? [\[Forests\]](#)
- (A) 300 years
 - (B) 60 years
 - (C) 65 years
 - (D) 360 years
 - (E) 240 years
 - (F) 12 years
 - (G) 120 years
- 68.** Find the value of $(-6) \times 4 - (-9) \times 2$. [\[Integers\]](#)
- (A) -42
 - (B) 42
 - (C) 6
 - (D) -12
 - (E) 12
 - (F) -6

69. Photosynthesis is a chemical change. A leaf makes 6 g of glucose and releases 6.4 g of oxygen using 9 g of water and some carbon dioxide. What mass of carbon dioxide did the leaf use? [\[Physical & Chemical Changes\]](#)

- (A) 3.4 g
- (B) 6.4 g
- (C) 9 g
- (D) 4.4 g
- (E) 2.4 g
- (F) 21.4 g

70. The breadth of a rectangle is $x - 2$. Its length is 4 more than twice the breadth. Find the simplest expression for the length. [\[Algebraic Expressions\]](#)

- (A) $2x - 4$
- (B) $x - 6$
- (C) $2x - 8$
- (D) $3x - 2$
- (E) $x + 2$
- (F) $2x + 4$
- (G) $2x$

71. A herd of 25 elephants each drops about 100 kg of dung a day, which returns nutrients to the forest. How many tonnes of dung enrich the forest over 30 days? (1 tonne = 1000 kg) [\[Forests\]](#)

- (A) 75 tonnes
- (B) 0.75 tonnes
- (C) 150 tonnes
- (D) 7.5 tonnes
- (E) 2500 tonnes
- (F) 750 tonnes
- (G) 3000 tonnes

72. On a number line, point A is at -20. Each step to the right is +7. Where is A after 4 steps to the right and then 1 step to the left? [\[Integers\]](#)

- (A) -13
- (B) 1
- (C) -1
- (D) 9
- (E) 15
- (F) 8
- (G) 5

73. A forest fire burns wood to ash. A 500 kg pile of dry wood leaves 30 kg of ash, and the rest leaves as smoke and gases. Burning is a chemical change. What mass leaves as smoke and gases? [\[Physical & Chemical Changes\]](#)

- (A) 470 kg
- (B) 250 kg
- (C) 485 kg
- (D) 500 kg
- (E) 440 kg
- (F) 30 kg

74. After simplifying $4(x + 2) - 3(x - 1)$, state the coefficient of x and the constant term. [\[Algebraic Expressions\]](#)

- (A) 1 and -11
- (B) -1 and 11
- (C) 4 and 8
- (D) 1 and 5
- (E) 1 and 11
- (F) 7 and 11
- (G) 7 and 5

75. Which one of these statements about forests is TRUE: forests increase floods, forests reduce rainfall, forests keep the balance of oxygen and carbon dioxide, or forests make soil infertile? [\[Forests\]](#)

- (A) forests reduce rainfall
- (B) forests keep the balance of oxygen and carbon dioxide
- (C) forests have no effect on air
- (D) forests increase carbon dioxide in air
- (E) forests increase floods
- (F) forests make soil infertile
- (G) forests dry up the soil

76. Two negative integers have a product of 36 and a sum of -13. What is the smaller (more negative) of the two integers? [\[Integers\]](#)

- (A) -9
- (B) -3
- (C) -6
- (D) -13
- (E) -12
- (F) -4
- (G) -36

77. Setting of cement is a chemical change that cannot be reversed. A builder uses 3 full bags of 50 kg each and 2 part-bags of 15 kg each. How many kilograms of cement does he use in all? [\[Physical & Chemical Changes\]](#)

- (A) 165 kg
- (B) 160 kg
- (C) 195 kg
- (D) 130 kg
- (E) 180 kg
- (F) 210 kg
- (G) 150 kg

78. A square has each side equal to $3a - 2$. Find the expression for its perimeter and its value when $a = 4$. [\[Algebraic Expressions\]](#)

- (A) 40
- (B) 10
- (C) 30
- (D) 48
- (E) 20
- (F) $12a - 2$

79. Producers in a forest hold 240000 kJ of energy. Only 10% passes to the next level at each step, through three steps up to the top carnivores. How much energy reaches the top carnivores? [\[Forests\]](#)

- (A) 24 kJ
- (B) 240 kJ
- (C) 2.4 kJ
- (D) 216000 kJ
- (E) 2400 kJ
- (F) 24000 kJ
- (G) 120 kJ

80. The midday temperatures for a week were -2, -4, 3, -1, 0, 5 and -8 degrees C. What was the average (mean) midday temperature for the week? [\[Integers\]](#)

- (A) -7 deg C
- (B) 1 deg C
- (C) -49 deg C
- (D) 0 deg C
- (E) -1 deg C
- (F) -2 deg C

81. Which one of these changes can be easily reversed: burning paper, freezing water into ice, cooking an egg, or the rusting of iron? [\[Physical & Chemical Changes\]](#)

- (A) cooking an egg
- (B) cooking an egg and rusting
- (C) burning paper
- (D) burning paper and cooking an egg
- (E) freezing water into ice
- (F) none of them
- (G) rusting of iron

82. By how much does the expression $6x - 5$ exceed the expression $2x + 3$ when $x = 7$? [\[Algebraic Expressions\]](#)

- (A) 4
- (B) 36
- (C) 8
- (D) 20
- (E) 50
- (F) 12
- (G) 28

83. A forest yields 8 kg of a useful medicinal herb from each hectare. A group harvests from 35 hectares but leaves one quarter of the area untouched so it can regrow. How many kilograms of herb do they harvest? [\[Forests\]](#)

- (A) 240 kg
- (B) 70 kg
- (C) 175 kg
- (D) 210 kg
- (E) 8.75 kg
- (F) 105 kg

84. Multiply -15 by the difference obtained when 5 is subtracted from -3. [\[Integers\]](#)

- (A) -30
- (B) 12
- (C) -12
- (D) 30
- (E) 120
- (F) 24
- (G) -120

85. Souring of milk, dissolving sugar in water, melting butter, and breaking a glass are four changes. Which one is a chemical change shown by a new substance forming? [\[Physical & Chemical Changes\]](#)

- (A) melting butter and breaking a glass
- (B) melting butter
- (C) none of them
- (D) dissolving sugar and breaking a glass
- (E) dissolving sugar in water
- (F) breaking a glass
- (G) souring of milk

86. If the value of $3x - 4$ is 11, find the value of $5x + 2$. [\[Algebraic Expressions\]](#)

- (A) 17
- (B) 7
- (C) 25
- (D) 37
- (E) 27
- (F) 13

87. A forest cooperative sells wild honey at 320 rupees per kilogram. It collects 4.5 kg of honey from each of its 60 hives. What is the total value of the honey collected? [\[Forests\]](#)

- (A) Rs 43200
- (B) Rs 8640
- (C) Rs 90000
- (D) Rs 1440
- (E) Rs 19200
- (F) Rs 86400

88. A diver starts at the surface (0 m). She descends 8 m, ascends 3 m, descends 6 m, and ascends 4 m. What is her final depth compared with the surface? [\[Integers\]](#)

- (A) -7 m
- (B) -11 m
- (C) 1 m
- (D) -1 m
- (E) 21 m
- (F) -21 m
- (G) 7 m

89. Digestion breaks food into simpler substances, which is a chemical change. If 250 g of food is eaten and 60% of it is absorbed as useful nutrients, what mass becomes undigested waste? [\[Physical & Chemical Changes\]](#)

- (A) 100 g
- (B) 150 g
- (C) 200 g
- (D) 60 g
- (E) 110 g
- (F) 40 g
- (G) 90 g

90. Subtract the sum of $2x - 3$ and $x + 5$ from $10x$. What expression do you get? [\[Algebraic Expressions\]](#)

- (A) $13x + 2$
- (B) $7x + 8$
- (C) $6x - 2$
- (D) $13x - 2$
- (E) $7x - 2$
- (F) $7x - 8$

91. Mushrooms and many bacteria feed on dead leaves, dead animals and droppings, breaking them down. To which group of living things in the forest do mushrooms and these bacteria belong? [\[Forests\]](#)

- (A) herbivores
- (B) green plants
- (C) decomposers
- (D) prey
- (E) carnivores
- (F) top carnivores

92. Evaluate the expression (-5) squared $+ (-5) \times 4$. [\[Integers\]](#)

- (A) 45
- (B) -45
- (C) -15
- (D) 15
- (E) -105
- (F) 5

93. Four iron nails are kept in: dry air, boiled water sealed under oil, ordinary tap water open to the air, and a dry sealed tube. Which nail rusts the most? [\[Physical & Chemical Changes\]](#)

- (A) the nail in dry air
- (B) the nail in boiled water under oil
- (C) all rust the same amount
- (D) the nail in boiled water under oil and in dry air
- (E) the nail in the dry sealed tube
- (F) the nail in tap water open to the air

94. A square tile has side $2x + 1$ and a rectangular tile has length $3x$ and breadth $x - 2$. Find how much greater the square tile's perimeter is than the rectangle's perimeter. [\[Algebraic Expressions\]](#)

- (A) 4
- (B) $2x - 4$
- (C) 6
- (D) $8x + 2$
- (E) $4x$
- (F) $2x + 4$
- (G) 2

95. When too many trees are cut down, several harmful effects follow together. Which set of effects is caused by such heavy deforestation? [\[Forests\]](#)

- (A) fewer floods and more rainfall
- (B) more oxygen and richer soil
- (C) more floods, soil erosion, less rainfall and more carbon dioxide
- (D) less carbon dioxide and more rain
- (E) steadier rivers and more rainfall
- (F) more humus and less erosion
- (G) cooler climate and more wildlife

96. At 6 pm the temperature was 4 degrees C, and it then dropped by 2 degrees C every hour. At what time did it first reach -10 degrees C? [\[Integers\]](#)

- (A) midnight
- (B) 7 pm
- (C) 5 am
- (D) 1 am
- (E) 2 am
- (F) 11 pm
- (G) 3 am

97. A kettle boils 500 g of water until it all turns to steam. The steam is captured and cooled, turning back into water. What mass of water is recovered, and what type of change is boiling and condensing? [\[Physical & Chemical Changes\]](#)

- (A) 500 g, chemical
- (B) 500 g, both kinds
- (C) 500 g, physical
- (D) 1000 g, physical
- (E) 450 g, physical
- (F) 0 g, chemical

98. If $m = -3$, find the value of the expression $4m$ squared $+ 2m - 7$. [\[Algebraic Expressions\]](#)

- (A) 35
- (B) 23
- (C) 47
- (D) 5
- (E) 11
- (F) -13
- (G) 29

99. A national park covers 4500 hectares. Two ninths of it is open grassland where deer graze, and the rest is dense forest. How many hectares are dense forest? [\[Forests\]](#)

- (A) 1500 hectares
- (B) 3000 hectares
- (C) 1000 hectares
- (D) 4000 hectares
- (E) 2500 hectares
- (F) 3500 hectares

100. On Mars, the temperature at dawn is -73 degrees C and by noon it rises to -19 degrees C. By how many degrees did the temperature rise from dawn to noon? [\[Integers\]](#)

- (A) -54 deg C
- (B) 46 deg C
- (C) -46 deg C
- (D) 88 deg C
- (E) 92 deg C
- (F) 54 deg C